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Comparison of Brain Structure Volumes in Insectivora and Primates.

V. Area striata (AS)

By

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With 10 Figures and 4 Tables

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Abstract and Summary: Volumes of the area striata (area 17, BRODMANN) were measured in 44 species of Primates and Scandentia and some of its laminar components in 25 species. The relative size (expressed by size indices) is on the average distinctly larger in simians than in isoponderous prosimians. The average dimensions of increase are 2.5 times for total area striata (ASV), 3.4 times for white matter (ASW), 2.35 times for grey matter (ASG), 1.9 times for lamina 1 (ASG1) and 2.4 times for laminae 2–6 (ASG2–6). The high increase of the white matter and the low increase of lamina 1 both correspond to similar results on the total neocortex (FRAHM et al., 1982). The molecular layer (ASG1) is on the average about 12% of the total area striata grey (ASG). The visual cortex thus has a narrower molecular layer than the neocortex as a whole, in which an average of 14% was found.

The size indices of man are within the simian range and generally slightly above the simian average. Tree shrews have relatively small visual cortices, but reach into the lower part of the prosimian range. For Insectivora, ASG volumes were estimated from CGL volumes. According to these estimates, the relative ASG size is about 1/90 in shrews, about 1/40 in tenrecs and about 1/10 in hedgehogs that of isoponderous prosimians. In the European hedgehog, the area striata is about 12% of the total neocortex, a value similarly to that found in area measurements by BRODMANN (1913). The relative size of the area striata is discussed with regard to functional, ecoethological and phylogenetic considerations.

With increasing body weight, the increase in the visual structures, both cortical and non-cortical, is weak when compared with other brain parts and body organs (strongly negative allometry).

Introduction

In contrast to previous parts of the current series (I. Neocortex, FRAHM et al., 1982; II. Accessory olfactory bulb, STEPHAN et al., 1982; III. Main olfactory bulb, BARON et al., 1983; IV. Non-cortical visual structures, STEPHAN et al., 1984) we were not able to include *measured* data from Insectivora in the comparisons of the present paper. The reason is that we were generally unable to determine homologous visual fields clearly in Insectivora with architectonic methods. In cases where this was possible, these fields could not be delimited with precision. Both clear homologization and delimitation are preconditions for exact measurement and well-founded comparisons. However, some information as to the size of the area striata (AS) in Insectivora could be obtained by plotting cortical versus non-cortical visual structures in Primates and by transposing existing interrelations to Insectivora.

The AS of Primates and Scandentia (tree shrews) is a well circumscribed structure which, in addition to common characteristics (stripe of Gennari, narrow and cell-poor fifth layer, narrow and cell-rich sixth layer), "has a considerable degree of variability in laminar pattern" (ZILLES et al., 1982). Despite this variability and less distinct characteristics in some lower prosimians, the grey matter of AS is always clear enough for delimitation. Thus, several investigators have made measurements and size comparisons of this field, which are lacking for other regions and fields of the neocortex. BRODMANN (1913) compared area measurements of the visual cortex in many species from several orders and included the hedgehog (*Erinaceus*) belonging to the order Insectivora. Area measurements in man were performed by BRODMANN (1911, 1913), TRAMER (1914, 1916), ECONOMO and KOSKINAS (1925), ECONOMO (1926), PUTNAM (1926), N. POPOFF (1927), I. POPOFF and N. POPOFF (1929) and FILIMONOFF (1932).

Volumetric measurements of the primary visual cortex in man were performed by GERHARDT and KREHT (1933), M. ROSE (1935) and STEPHAN (1969). STEPHAN gave a concise comparison of AS volumes in 41 primate species.

The present publication is a continuation of this latter work. It contains measurements on some more species, distinctly more specimens and two laminar components (lamina 1, laminae 2–6) of AS. The relative size of AS and its components is discussed in relation to (1) that of non-cortical visual structures and neocortex, and (2) ecology and behaviour of various systematic units.

Material and Methods

The volume of the total area striata was measured in 67 individuals from 44 species belonging to Primates (18 species of prosimians and 23 species of simians) and Scandentia (3 species). In 25 of the 44 species, white matter, grey matter, and within the latter, lamina 1 and laminae 2–6 were measured separately. Details on the material and the methods of measurements were described by STEPHAN et al. (1981).

With regard to the delimitation of the white matter of the area striata (ASW), it must be stressed that its borders are in general artificially defined by the shortest distance between the borders of the grey matter in microphotographs of the histological sections. Such delimitation of course gives no exact data on ASW volume. However, exact data can hardly be obtained when limits are traced across the white matter. Thus the results of the measurements of the ASW and its comparison can be evaluated only with reservation. The data of the total area striata (which includes the ASW) are influenced also by this inaccuracy and, therefore, size comparisons of the primary visual cortex are focussed on the grey matter.

Volume measurements are rather time consuming. In order to obtain a broader spectrum of species for comparison, we refrained from investigating more than 1–2 individuals per species. In a few species only more than 2 individuals were investigated to provide some insight into variability. All the data were obtained from photographs of serial sections and corrected to fresh volume (Table 1).

Sizes were compared using three different methods: (1) **percentage values and/or ratios**: Percentages were preferred when the given structure is included in the reference system, e.g. area striata in % of the neocortex, whereas ratios were preferred when two different structures were compared, e.g. area striata and corpus geniculatum laterale; (2) **interstructural regression** line analyses between structures; and (3) **size indices (SI)**, which evaluate size differences of the given structures in isoponderous species. This method determines the interdependency between body weight and size (weight, volume, surface area) of the brain structure by calculating the respective regression lines within narrowly related groups. In double logarithmic scales the regression lines are straight lines with a rather uniform slope, which indicates the rate at which the brain structure enlarges with increasing body weight. Much care was given to the estimation of the slopes of the reference lines. The method will be demonstrated for the total area striata (ASV). The slopes of the regression lines (ASV versus BoW) of the eight primate families under investigation (with more than one species) had an average

of 0.532. The regression line of all prosimian species ($n 18$) had a slope of 0.513 and of all non-human simians ($n 22$), a slope of 0.551. The intermediate value between these two was also 0.532. When NV is plotted against BoW (part I, FRAHM et al., 1982), a slope of 0.67 is obtained. Combined with the slope of the interstructural regression of ASV versus NV (0.787, see later), the expected slope for ASV versus BoW was 0.527 ($0.787 \cdot 0.67$). This value is consistent with the above results on regression line analysis between ASV and BoW in narrowly related groups. The average of the three values (0.532, 0.532, 0.527) was 0.530 and therefore, the reference line for the calculation of size indices (Fig. 1) was given a slope of 0.53. Similar analyses resulted in 0.68 for the white matter (ASW), and in 0.51 for the grey matter (ASG) and its laminar components (ASG1 and ASG2–6).

Reference lines with these gradients were drawn through the average log structural volume and the average log body weight of the prosimians (in Fig. 1 for ASV). The distances of the various species from the reference lines are expressed in the form of size indices and given in percentages. All points on the line have 100%.

Results

The measurements on area striata and four of its components in 25 species allow for so many comparisons that it is impracticable to mention all of them. To allow for detailed comparisons between species, we give the raw data in Table 1, but confine ourselves to a discussion of substantial differences and some general trends.

Total area striata

Percentage: The percentage values of area striata volume (ASV) relative to (1) brain weight (BrW), (2) telencephalon volume (TV), and (3) neocortex volume (NV) are given in Table 2, columns 1–3. To obtain a general view, averages for several systematic groups are also given.

Generally, the percentage values were low in *Homo*, Pongidae and in *Daubentonina*, and high in *Tarsius*, Cheirogaleidae and Callitrichidae. In man, the very low percentages were probably influenced mainly by a very large non-visual neocortex; in *Tarsius* the rather high percentages may be the consequence of a very large AS. However, even in these species with extreme values, it is impossible to make firm statements about the predominating influence.

Similarly the question of an evolutionary trend in the size of AS cannot be answered by percentage values. The low values of ASV percentages in higher primates and man in columns 1–3 of Table 2 may create the impression that AS size generally decreases in higher primates. This is in contradiction to the results based on comparisons of size indices.

Interstructural regression: When the ASV was plotted against NV in a double logarithmic scale, the overall regression line ($n 44$) had a slope of 0.787 and

Table 1. Volumes of area striata and some of its laminar components (in mm³).

	n	total volume*	white matter	grey matter	lamina 1	laminae 2-6
	(1)	ASV (2)	ASW (3)	ASG (4)	ASG 1 (5)	ASG 2-6 (6)
Scandentia						
<i>Tupaia glis</i>	4	179	10.2	169	19.6	149
<i>Tupaia minor</i>	2	164	12.5	152	—	—
<i>Urogale everetti</i>	1	136	12.7	123	15.5	108
Primates						
Prosimians						
<i>Microcebus murinus</i>	5	144	8.24	136	15.0	121
<i>Cheirogaleus major</i>	2	477	44.8	433	—	—
<i>Cheirogaleus medius</i>	2	216	17.6	199	—	—
<i>Lemur albifrons</i> ¹⁾	2	1,519	267	1,253	138	1,114
<i>Varecia variegata</i> ²⁾	1	2,112	367	1,745	—	—
<i>Lepilemur ruficaudatus</i>	3	358	31.4	326	52.5	274
<i>Avahi laniger laniger</i>	1	542	66.5	475	—	—
<i>Avahi lan. occidentalis</i>	2	527	77.2	450	64.6	385
<i>Propithecus verreauxi</i>	2	1,355	249	1,106	131	974
<i>Indri indri</i>	2	1,961	313	1,648	279	1,369
<i>Daubentonia madagascar.</i>	1	1,355	242	1,113	189	923
<i>Loris tardigradus</i>	1	588	59.6	529	—	—
<i>Nycticebus coucang</i>	1	828	132	696	93.8	603
<i>Perodicticus potto</i>	2	552	57.7	495	—	—
<i>Galago senegalensis</i>	1	334	32.6	301	—	—
<i>Otolemur crassicaudatus</i> ³⁾	1	548	68.4	479	—	—
<i>Galagoides demidoff</i> ⁴⁾	2	237	22.4	214	25.6	189
<i>Tarsius spp.</i>	2	370	21.3	349	41.2	307
Simians						
<i>Callithrix jacchus</i>	2	685	119	566	57.8	508
<i>Cebuella pygmaea</i>	1	415	51.4	363	29.7	334
<i>Saguinus midas</i> ⁵⁾	1	1,061	220	841	—	—
<i>Saguinus oedipus</i>	1	977	170	807	—	—
<i>Cebus spp.</i>	2	4,703	939	3,764	365	3,399
<i>Aotus trivirgatus</i>	2	1,203	207	996	113	883
<i>Callicebus moloch</i>	1	1,524	324	1,201	—	—
<i>Saimiri sciureus</i>	1	2,326	442	1,884	175	1,709
<i>Pithecia monachus</i>	1	2,130	451	1,679	—	—
<i>Alouatta sp.</i>	1	2,461	543	1,918	220	1,698
<i>Ateles geoffroyi</i>	1	4,738	985	3,753	—	—
<i>Lagothrix lagothricha</i>	1	6,082	1,226	4,856	604	4,252
<i>Macaca mulatta</i>	1	6,586	1,243	5,343	—	—
<i>Cercocebus albigena</i>	1	6,831	1,411	5,420	—	—
<i>Papio anubis</i>	1	13,741	3,740	10,001	919	9,082
<i>Cercopithecus ascanius</i>	1	5,106	834	4,272	416	3,855
<i>Cercopithecus mitis</i>	1	5,275	1,042	4,232	—	—
<i>Miopithecus talapoin</i> ⁶⁾	1	3,048	709	2,339	—	—
<i>Colobus badius</i>	2	3,979	926	3,053	360	2,693
<i>Hylobates lar</i>	1	5,750	1,176	4,574	562	4,012
<i>Pan troglodytes</i>	1	14,597	4,004	10,593	1,156	9,437
<i>Gorilla gorilla</i>	1	15,185	3,123	12,062	—	—
<i>Homo sapiens</i>	1	20,226	5,439	14,787	1,704	13,083

Classification and species names are adapted to the list given by CORBET and HILL (1980). Different names used in former papers and in STEPHAN et al. (1981) are: 1) *Lemur fulvus*; 2) *Lemur variegatus*; 3) *Galago crassicaudatus*; 4) *Galago demidovii*; 5) *Saguinus tamarin*; 6) *Cercopithecus talapoin* (for Tables 1-3) and 7) *Aethechinus algirus*; and 8) *Crocidura occidentalis* (for Table 4).

* Most of the area striata volumes deviate slightly from those given in the data collected by STEPHAN et al. (1981) because of a broadening of the material and/or re-investigation in more highly-enlarged photographic series.

Table 2. Percentages of total area striata (ASV) and some of its laminar components (ASW = white matter; ASG = grey matter; ASG1 = lamina 1; ASG 2-6 = laminae 2-6) relative to brain weight (BrW), telencephalon volume (TV), neocortex (NV) and its components (NW, NG, NG1, NG 2-6), and ratios between ASG and dorsal nucleus of corpus geniculatum laterale (CGL).

			ASV	ASV	ASV	ASW	ASG	ASG1	ASG2-6	ASG	ASG1	Ratio
			BrW*)	TV*)	NV*)+)	NW+)	NG+)	NG1+)	NG2-6+)	ASV	ASG	ASG
			(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Scandentia												
			5.6	9.6	17.3	10.8	17.9	15.4	18.3	94.3	11.6	13.1
			6.4	11.3	21.2	—	—	—	—	92.4	—	15.4
			3.2	5.6	10.2	6.4	10.8	9.8	11.0	90.6	12.6	12.3
Primates (Prosimians)												
			8.1	12.8	19.5	12.6	20.2	15.8	20.9	94.3	11.0	19.2
			7.0	11.1	16.2	—	—	—	—	90.6	—	20.1
			6.9	11.1	17.7	—	—	—	—	91.9	—	18.4
			6.5	9.8	12.4	8.8	13.7	10.7	14.2	82.5	11.0	24.2
			6.7	10.3	13.8	—	—	—	—	82.6	—	25.3
			4.7	7.6	10.9	6.6	11.6	10.7	11.8	91.2	16.1	20.9
			5.2	8.3	11.3	—	—	—	—	87.7	—	17.2
			5.4	8.7	11.9	9.9	12.3	11.1	12.5	85.3	14.4	15.4
			5.1	8.0	10.3	7.5	11.2	9.2	11.6	81.6	11.9	17.2
			5.1	7.7	9.7	7.4	10.4	10.3	10.4	84.0	16.9	17.4
			3.0	4.5	6.1	4.8	6.5	6.9	6.4	82.2	17.0	19.0
			8.9	12.9	16.7	—	—	—	—	89.9	—	21.5
			6.6	9.7	13.4	10.6	14.1	13.2	14.2	84.1	13.5	19.4
			3.9	5.9	8.3	—	—	—	—	89.6	—	20.4
			7.0	11.1	15.6	—	—	—	—	90.2	—	18.8
			5.3	8.4	11.6	—	—	—	—	87.5	—	20.7
			7.0	10.6	15.1	14.9	15.1	12.7	15.5	90.6	12.0	19.7
			10.3	16.4	20.9	15.5	21.4	17.4	22.0	94.2	11.8	16.7
Primates (Simians)												
			9.0	12.9	15.7	14.7	15.9	13.1	16.3	82.6	10.2	22.6
			9.2	13.4	16.4	13.0	17.0	13.3	17.4	87.6	8.2	21.5
			10.3	15.0	18.0	—	—	—	—	79.3	—	23.4
			9.8	13.9	16.6	—	—	—	—	82.6	—	23.6
			6.6	9.0	10.1	5.9	12.3	10.1	12.6	80.0	9.7	27.5
			7.0	9.9	12.1	9.5	12.8	10.9	13.1	82.8	11.4	30.3
			8.0	11.3	13.7	—	—	—	—	78.8	—	22.1
			9.7	13.2	15.0	11.9	15.9	13.8	16.2	81.0	9.3	30.0
			6.1	8.5	10.1	—	—	—	—	78.8	—	22.0
			4.7	6.6	7.8	5.5	8.8	7.5	9.0	77.9	11.5	21.9
			4.4	5.9	6.7	—	—	—	—	79.2	—	24.9
			6.0	8.1	9.2	7.3	9.9	7.9	10.3	79.9	12.4	29.6
			7.1	9.3	10.4	—	—	—	—	81.1	—	33.7
			6.6	8.9	9.9	—	—	—	—	79.3	—	29.8
			6.8	8.9	9.8	6.9	11.7	8.6	12.1	72.8	9.2	25.4
			7.6	10.0	11.3	6.4	13.3	10.4	13.7	83.7	9.8	29.1
			7.0	9.4	10.6	—	—	—	—	80.2	—	28.3
			7.6	10.1	11.5	—	—	—	—	76.7	—	21.4
			5.1	6.9	7.8	6.5	8.3	7.0	8.6	76.7	11.8	23.8
			5.6	7.6	8.7	6.1	9.8	8.9	9.9	79.6	12.3	26.1
			3.6	4.7	5.0	3.4	6.1	5.3	6.3	72.6	10.9	29.8
			3.0	4.1	4.5	—	—	—	—	79.4	—	31.4
			1.5	1.9	2.0	1.3	2.5	2.3	2.6	73.1	11.5	35.6
(Means)												
	n	n										
overall	44	(25)	6.4	9.3	12.1	8.6	12.4	10.5	12.7	83.7	11.9	22.9
tree shrews	3	(2)	5.1	8.8	16.2	8.6	14.4	12.6	14.7	92.4	12.1	13.6
primates	41	(23)	6.5	9.4	11.8	8.6	12.2	10.3	12.5	83.1	11.9	23.5
prosimians	18	(10)	6.3	9.7	13.4	9.9	13.7	11.8	14.0	87.8	13.6	19.5
simians	23	(13)	6.6	9.1	10.6	7.6	11.1	9.2	11.4	79.4	10.6	26.7
marmosets	4	(2)	9.6	13.8	16.7	13.9	16.5	13.2	16.9	83.0	9.2	22.8
monkeys	15	(8)	6.7	9.1	10.4	7.5	11.6	9.5	12.0	79.3	10.6	26.7
apes	3	(2)	4.1	5.5	6.0	4.8	8.0	7.1	8.1	77.2	11.6	29.1
man	1	(1)	1.5	1.9	2.0	1.3	2.5	2.3	2.6	73.1	11.5	35.6

Data taken from: *) STEPHAN et al. (1981); +) FRAHM et al. (1982); §) STEPHAN et al. (1984).

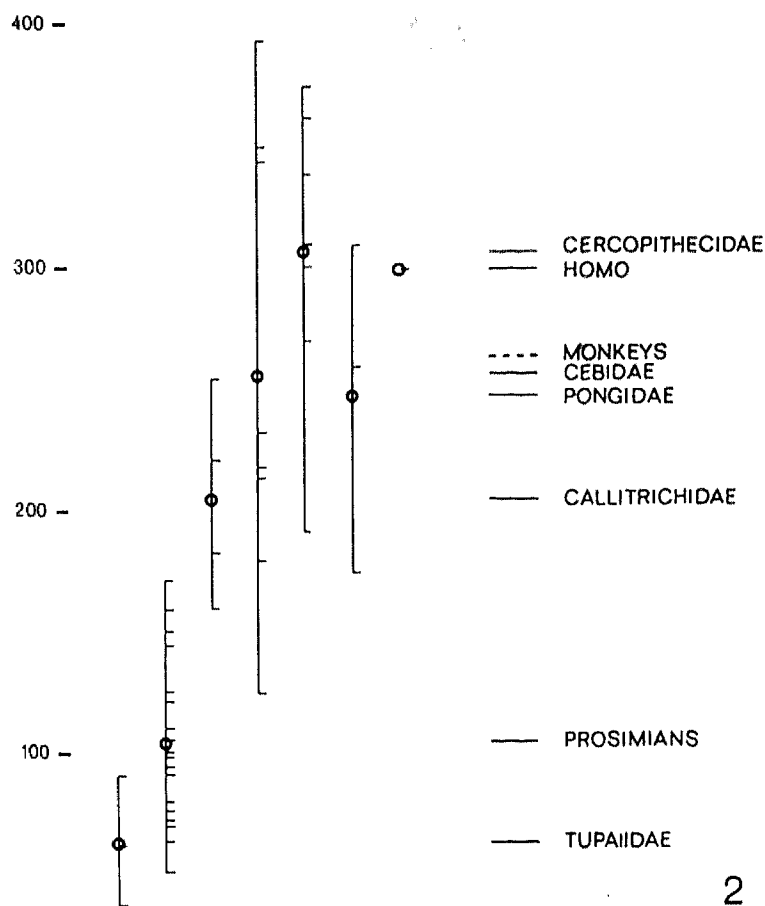
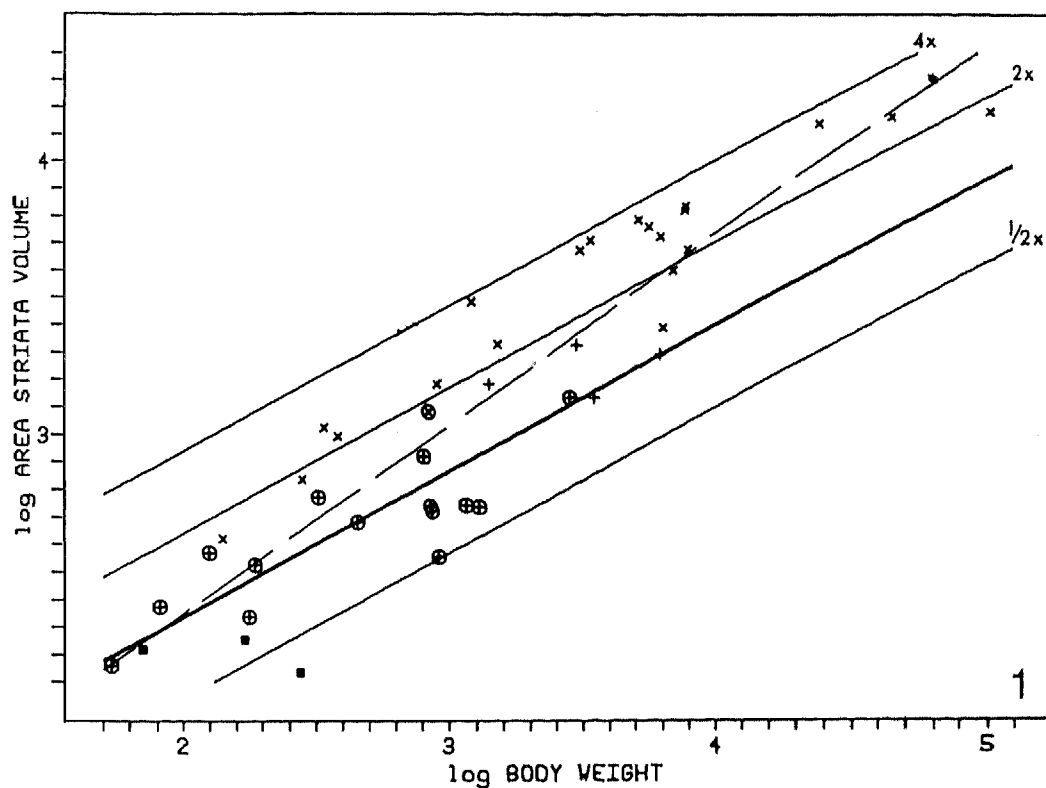


Fig. 1. Area striata volume (ASV) plotted against body weight (BoW) in a double logarithmic scale. The volumes are in mm^3 , the weights in grams. The thick reference line is through the average ASV and average BoW of the prosimians with the common slope found for the primate families and suborders (see text). Its formula is $\log \text{ASV} = 1.278 + 0.53 \times \log \text{BoW}$. The thinner parallel lines are in multiples of 2 and refer to the size indices ($\times 100$) given in Table 3 and scaled in Figure 2. ■ tree shrews; + prosimians; x simians, * man. Nocturnal primates are encircled.

Fig. 2. Size indices of area striata volumes. The indices are numerical manifestations of the distances from the thick line given in Fig. 1, and are given in their percentage form ($\times 100$). Each of the small horizontal bars represents the size index for a species. The range (vertical bars) and the mean (○) of various groups are given on the left, and the sequences of means for the various groups are presented on the right.

Table 3. Size indices of total area striata (ASV), and some of its laminar components (ASW = white matter; ASG = grey matter; ASG1 = lamina 1; ASG2-6 = laminae 2-6). The indices reflect the distances from the reference lines: $\log \text{ASV} = 1.278 + 0.53 \log \text{BoW}$; $\log \text{ASW} = -0.085 + 0.68 \log \text{BoW}$; $\log \text{ASG} = 1.277 + 0.51 \log \text{BoW}$; $\log \text{ASG1} = 0.412 + 0.51 \log \text{BoW}$; $\log \text{ASG2-6} = 1.227 + 0.51 \log \text{BoW}$.

	ASV (1)	ASW (2)	ASG (3)	ASG1 (4)	ASG2-6 (5)
Scandentia					
<i>Tupaia glis</i>	62	38	65	55	64
<i>Tupaia minor</i>	91	84	92		
<i>Urogale everetti</i>	37	34	37	34	36
Primates					
Prosimians					
<i>Microcebus murinus</i>	92	66	94	75	94
<i>Cheirogaleus major</i>	99	86	101		
<i>Cheirogaleus medius</i>	73	63	75		
<i>Lemur albifrons</i>	172	235	165	132	164
<i>Varecia variegata</i>	160	192	155		
<i>Lepilemur ruficaudatus</i>	51	37	53	62	50
<i>Avahi laniger laniger</i>	64	62	65		
<i>Avahi laniger occidentalis</i>	77	95	76	79	73
<i>Propithecus verreauxi</i>	95	118	91	79	90
<i>Indri indri</i>	101	100	101	124	94
<i>Daubentonia madagascariensis</i>	106	133	103	127	96
<i>Loris tardigradus</i>	145	143	147		
<i>Nycticebus coucang</i>	126	170	122	119	118
<i>Perodicticus potto</i>	70	58	72		
<i>Galago senegalensis</i>	111	114	111		
<i>Otolemur crassicaudatus</i>	81	85	81		
<i>Galagoides demidoffi</i>	122	137	120	104	119
<i>Tarsius spp.</i>	151	97	157	134	155
Simians					
<i>Callithrix jacchus</i>	183	313	169	125	170
<i>Cebuella pygmaea</i>	160	217	155	92	159
<i>Saguinus midas</i>	255	508	228		
<i>Saguinus oedipus</i>	221	363	206		
<i>Cebus spp.</i>	350	482	330	232	334
<i>Aotus trivirgatus</i>	180	261	171	141	170
<i>Callicebus moloch</i>	219	386	198		
<i>Saimiri sciureus</i>	393	649	363	245	370
<i>Pithecia monachus</i>	233	379	213		
<i>Alouatta sp.</i>	125	170	116	96	115
<i>Ateles geoffroyi</i>	214	265	203		
<i>Lagothrix lagothricha</i>	344	443	327	294	321
<i>Macaca mulatta</i>	301	341	292		
<i>Cercocebus albigena</i>	310	383	295		
<i>Papio anubis</i>	339	464	302	201	308
<i>Cercopithecus ascanius</i>	362	402	357	252	362
<i>Cercopithecus mitis</i>	270	330	258		
<i>Miopithecus talapoin</i>	375	694	333		
<i>Colobus badius</i>	192	273	177	151	175
<i>Hylobates lar</i>	310	399	294	261	289
<i>Pan troglodytes</i>	260	328	235	186	234
<i>Gorilla gorilla</i>	175	146	175		
<i>Homo sapiens</i>	300	353	274	229	272

Table 3 (Continued)

			ASV (1)	ASW (2)	ASG (3)	ASG1 (4)	ASG2-6 (5)
Means	n	n					
overall	44	(25)	185	243	176	145	177
tree shrews	3	(2)	63	52	65	45	50
primates	41	(23)	194	257	184	154	188
prosimians	18	(10)	105	111	105	104	105
simians	23	(13)	264	372	247	193	252
marmosets	4	(2)	205	350	190	109	165
monkeys	15	(8)	280	395	262	202	269
apes	3	(2)	248	291	235	224	262
man	1	(1)	300	353	274	229	272

a correlation coefficient (r^2) of 0.967. The slope (distinctly smaller than 1) indicates that with increasing neocortex, the increase of AS is distinctly lower.

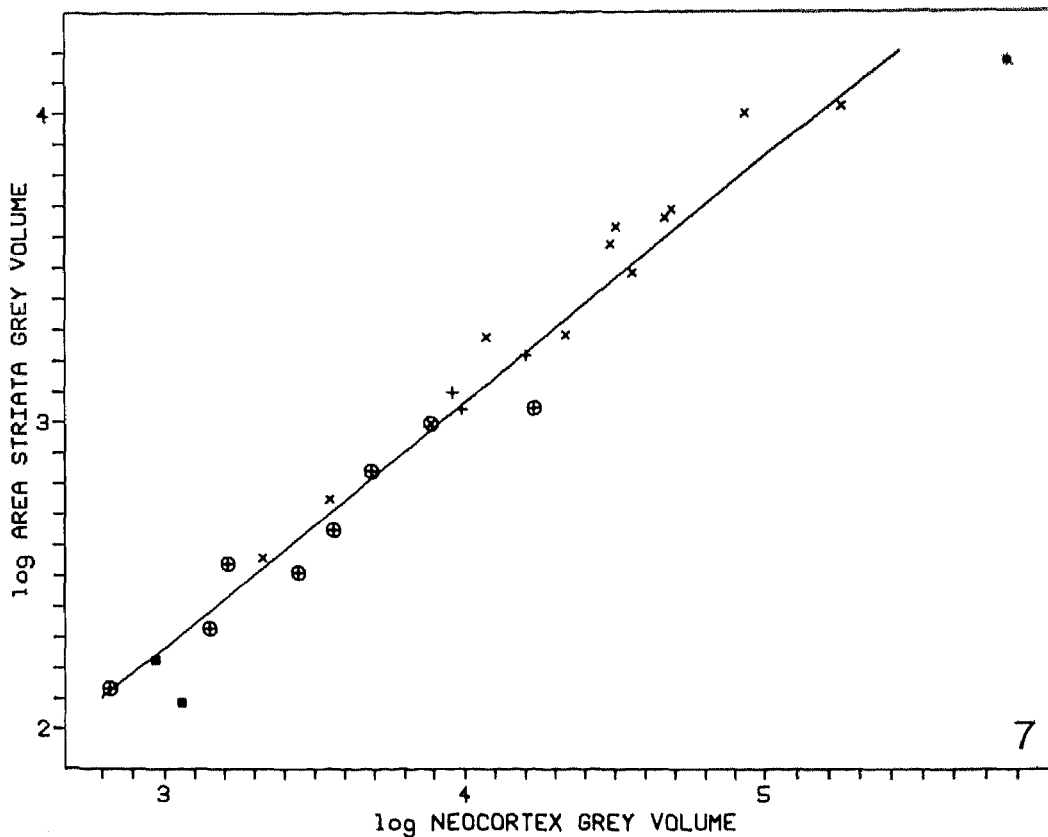
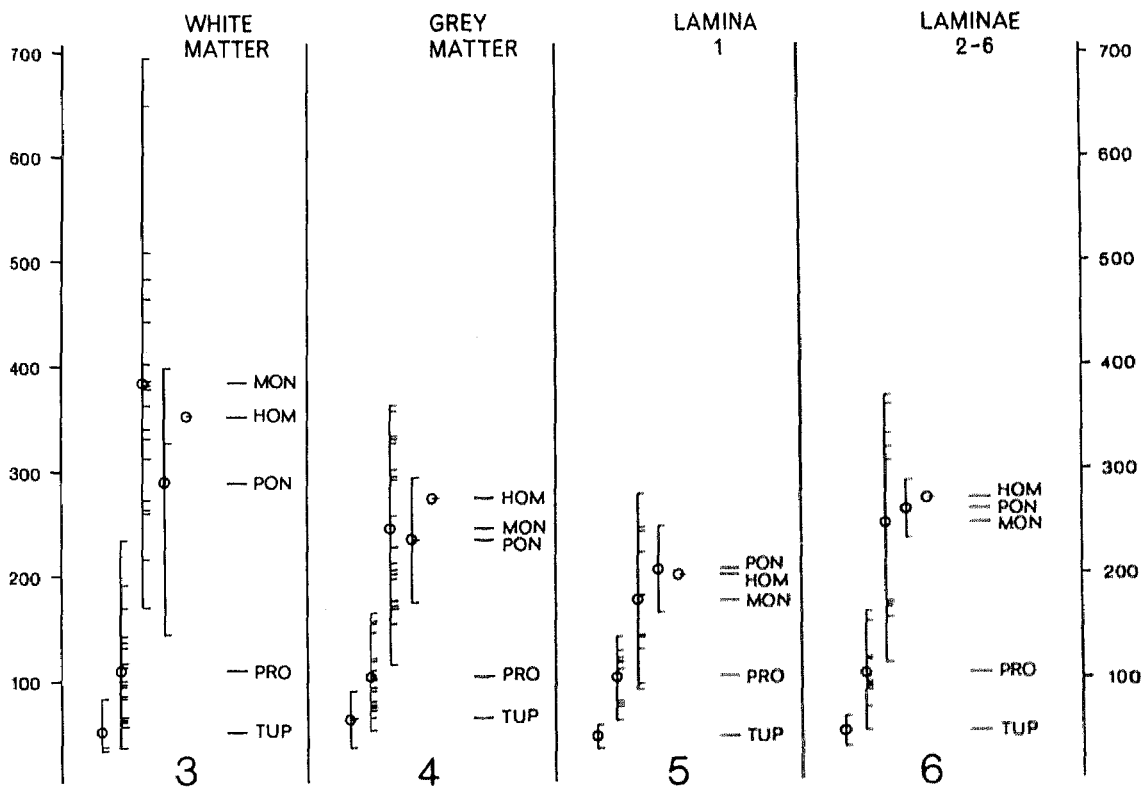
Size indices: The size indices were based on the reference line drawn through the average log ASV volume and the average log body weight of the prosimians with the slope of 0.53 (Fig. 1). The indices are given in Table 3, column 1, and scaled in Figure 2. In contrast to the percentage values and the inter-structural comparisons, they are independent of other brain structures. The indices showed a great variability within primates, both prosimian and simian. Since the ASV indices may be influenced by the inaccuracies of the ASW measurements, the comparisons are mainly based on the ASG indices (see later) which are similar (cf. columns 1 and 3 in Table 3 and Figs. 2 and 4).

White matter

In spite of the difficulties in delimitation (see Material and Methods section) the ASW data are presented for several reasons: (1) to possibly recognize general trends and (2) to enable comparisons with similar measurements of the white matter of the total neocortex (NW).

Percentage: The percentages of ASW relative to NW showed a great variability within systematic groups, but besides the very low value for man, there was no clear difference in the average percentage values between prosimians, simians and tree shrews.

The percentages of ASW relative to ASV complement the values for ASG relative to ASV (Table 2, column 8) bringing them to 100. Thus, to obtain the ASW to ASV values, the ASG to ASV percentage must be subtracted from 100. These values also had a high variability within the various systematic groups but nevertheless showed clearly larger averages (1) in primates compared with tree shrews and (2) in higher primates compared with lower ones. There was a strong tendency within the total AS



Figs. 3–6. Size indices of area striata components: white matter (Fig. 3), grey matter (Fig. 4), lamina 1 (Fig. 5), and laminae 2–6 (Fig. 6). HOM: *Homo sapiens*; MON: Monkeys; PON: Pongidae; PRO: Prosimians; TUP: Tupaia (Tree shrews). For further explanations see Figure 2.

Fig. 7. Volume of area striata grey (ASG) plotted against volume of neocortex grey (NG) in a double logarithmic scale. All volumes are in mm³. The formula for the regression line is $\log ASG = -0.1274 + 0.7974 \cdot \log NG$. The correlation coefficient (r^2) is 0.962. Symbols as in Figure 1.

for an increase in the white matter from lower to higher primates, and accordingly a decrease of the grey matter.

Interstructural regression: The increasing percentage of ASW relative to ASV can also be demonstrated by plotting the volumes on a double logarithmic scale (Fig. 8). The overall regression line (n 44) had a slope of 1.273, which is clearly larger than 1. The correlation coefficient (r^2) was 0.985. When plotting ASW directly against ASG, the slope must become still higher and was in fact found to be 1.331. The correlation coefficient (r^2) was 0.980.

When plotting ASW versus NW the slope was 0.816 (n 25) with a correlation coefficient (r^2) of 0.970. The slope indicates that with increasing NW, the increase of ASW is distinctly lower. This is in accordance with the results found for the total structures (ASV versus NV).

Size indices: The size indices of ASW are given in Table 3, column 2, and are scaled in Figure 3. The reference line for the calculation of size indices for ASW has a slope of 0.68. This slope is relatively high when compared with that for ASV plotted against BoW (0.68 versus 0.53). With increasing body weight (and hence brain weight), the increase in white matter is distinctly greater than the increase in grey matter, thus confirming results found for the total neocortex (see FRAHM et al., 1982).

Grey matter

Percentage: The percentages of ASG relative to NG are given in Table 2, column 5. There was a great variability of the values within the various systematic groups, and the means showed a slight tendency to decrease especially in the higher primates. In primates on the average about 1/8 (12.2%) of NG is ASG, whereas in man this proportion is only 1/40 (2.5%).

The percentages of ASG relative to ASV are given in Table 2, column 8. Since the values complement those of the white matter to 100, the averages are complementary also, and the statements made with regard to the white matter (see above) are reversed. The data point to a percentage decrease of ASG relative to ASV in the higher primates and man.

Interstructural regression: When the volumes of ASG were plotted against ASV in a double logarithmic scale (Fig. 8), the overall regression line had a slope of 0.953, which is distinctly smaller than 1. The very high correlation coefficient (r^2 0.9994) is not surprising since on the average, more than 4/5 (83%) of ASV is ASG.

In a plot of ASG versus NG (Fig. 7) the slope was 0.797 (n 25), with a correlation coefficient (r^2) of

0.962. The slope (clearly smaller than 1) indicates that with increasing NG, the increase of ASG is distinctly lower. Similar results were found for ASV and ASW relative to NV and NW. A few species deviated quite strongly from the regression line of Figure 7. These were *Urogale everetti*, *Daubentonia madagascariensis* and *Homo sapiens*, all of which were clearly below the line. In man, the neocortex is extraordinary large (very high index) and within the prosimians, *Daubentonia* also has the largest index (FRAHM et al., 1982). The size of the neocortex of *Urogale* is low within Scandentia. Thus, for this last species, the development of the visual cortex must be still lower (see below), whereas for *Homo sapiens* and *Daubentonia*, the large neocortex may have a greater influence on the deviation from the regression line.

When plotting CGL versus ASG (Fig. 10), the slope was 0.848 (n 44). This slope, distinctly smaller than 1, indicates that the increase of ASG clearly exceeds that of CGL. The correlation coefficient was relatively high (r^2 0.985), thus indicating that there is a close size relationship between the two structures. This close interrelationship accounts for all the systematic groups so far investigated and thus may be valid also for Insectivora. Therefore it will be used to estimate the size of the AS in this order (see Discussion).

Size indices: The indices of ASG are given in Table 3, column 3 and are scaled in Figure 4. The ASG slope is somewhat lower than that of ASV plotted against BoW (0.51 versus 0.53), in accordance with the high slope of ASW. With increasing body weight (and hence brain weight), the increase in ASG is distinctly lower than the increase in ASW, thus confirming the finding for the total neocortex (see FRAHM et al., 1982).

The ASG indices were only slightly different from the ASV indices. The lowest ASG indices were found in one of the tree shrews (*Urogale everetti*, SI 37) and within Primates, in the prosimian *Lepilemur ruficaudatus* (53). The highest ASG indices were found in *Saimiri sciureus* and *Cercopithecus ascanius* (363 and 357). The ratio of lowest and highest values indicates that in *Saimiri* and *C. ascanius*, the ASG is nearly seven times larger than in *Lepilemur* of equal body weight. The indices showed a great variability both in prosimians and simians.

Within prosimians, the ASG indices ranged from 53 in *Lepilemur* to 165 in *Lemur albifrons*. Both species belong to the same family, Lemuridae, and thus, within this family, the ASG may be 3.1 times larger in one species than in another. Average ASG indices of other prosimian families were 90 for Cheirogaleidae, 83 for Indridae and 109 for Lorisidae. The

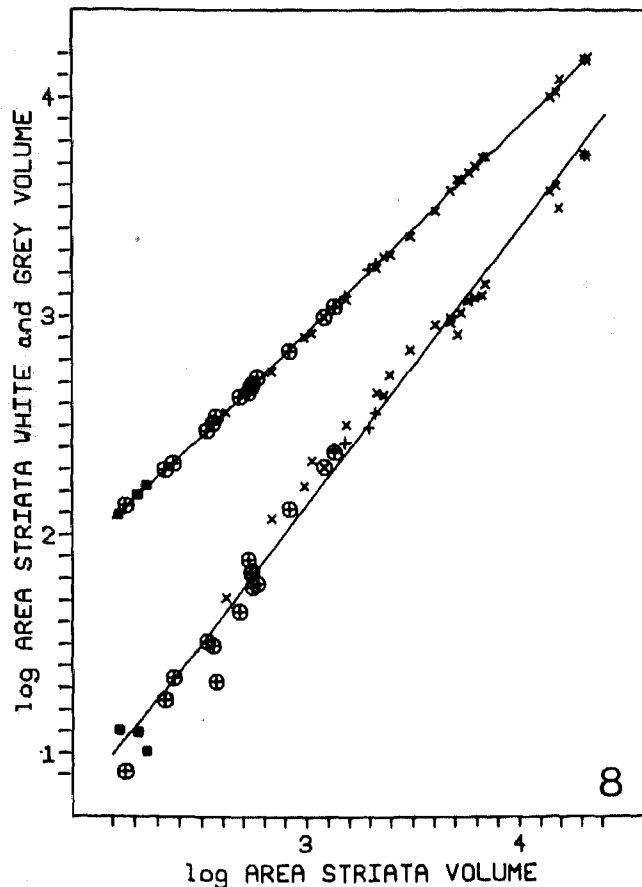


Fig. 8. Volumes of area striata white and grey matters (ASW and ASG) plotted against volume of total area striata (ASV) in a double logarithmic scale. All volumes are in mm³. Symbols as in Figure 1. Formulae for the regression lines are:

$$\log ASW = -1.680 + 1.273 \cdot \log ASV \quad (r^2 \ 0.985) \text{ for white matter (lower line) and}$$

$$\log ASG = 0.068 + 0.953 \cdot \log ASV \quad (r^2 \ 0.9994) \text{ for grey matter (upper line).}$$

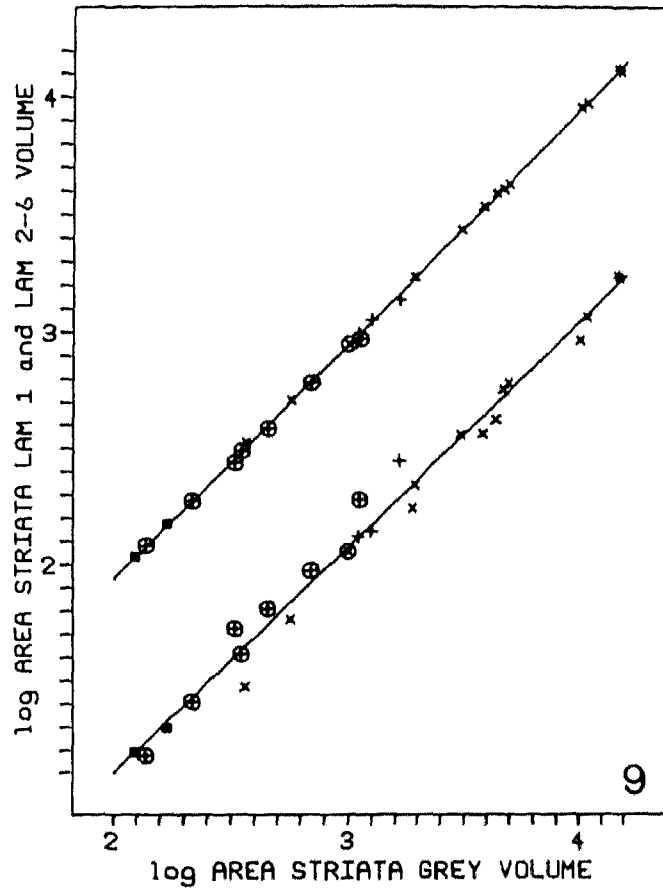


Fig. 9. Volumes of area striata molecular layer (ASG1) and area striata cell layers (ASG2-6) plotted against volume of area striata grey (ASG) in a double logarithmic scale. All volumes are in mm³. Symbols as in Figure 1. Formulae for the regression lines are:

$$\log ASG1 = -0.849 + 0.973 \cdot \log ASG \quad (r^2 \ 0.983) \text{ for lamina 1 (lower line) and}$$

$$\log ASG2-6 = -0.067 + 1.004 \cdot \log ASG \quad (r^2 \ 0.9997) \text{ for laminae 2-6 (upper line).}$$

average values for simian families were 190 for Callitrichidae, 240 for Cebidae, 288 for Cercopithecidae and 235 for Pongidae. Within each of these families, there were one or more strongly divergent species. The average ASG index of prosimians was close to 100 (which is the reference basis) and the average of simians was 247. Thus, the primary visual cortex of simians is on the average 2.5 times larger than that of isoponderous prosimians.

The index of man was slightly above the simian average (Fig. 4). This confirms the suggestion that in man the deviation from the ASG versus NG regression line (Fig. 7) is due to an especially large neocortex and not to an especially small AS. The same accounts for *Daubentonia*, whose ASG index (103) was close to the prosimian average. In contrast, the low position of *Urogale* in Figure 7 results mainly

from a small AS. In *Urogale*, the lowest ASG index (37) of all species so far investigated was found. Its NG index, when related to the average of prosimians, would be 46 (676:1466; values taken from FRAHM et al., 1982), and thus somewhat higher than the ASG index. Both the neocortex and AS are poorly developed in *Urogale*. This results not only from a comparison with average prosimians, but also with *Tupaia glis*, another representative of the order Scandentia in our material.

Lamina 1 of area striata grey (ASG 1)

Percentage: The percentage values of the area striata molecular layer (ASG 1) relative to ASG are given in Table 2, column 9. The values point to a slight percentage decrease of ASG 1 in simians. Such a decrease corresponds with the results found for the

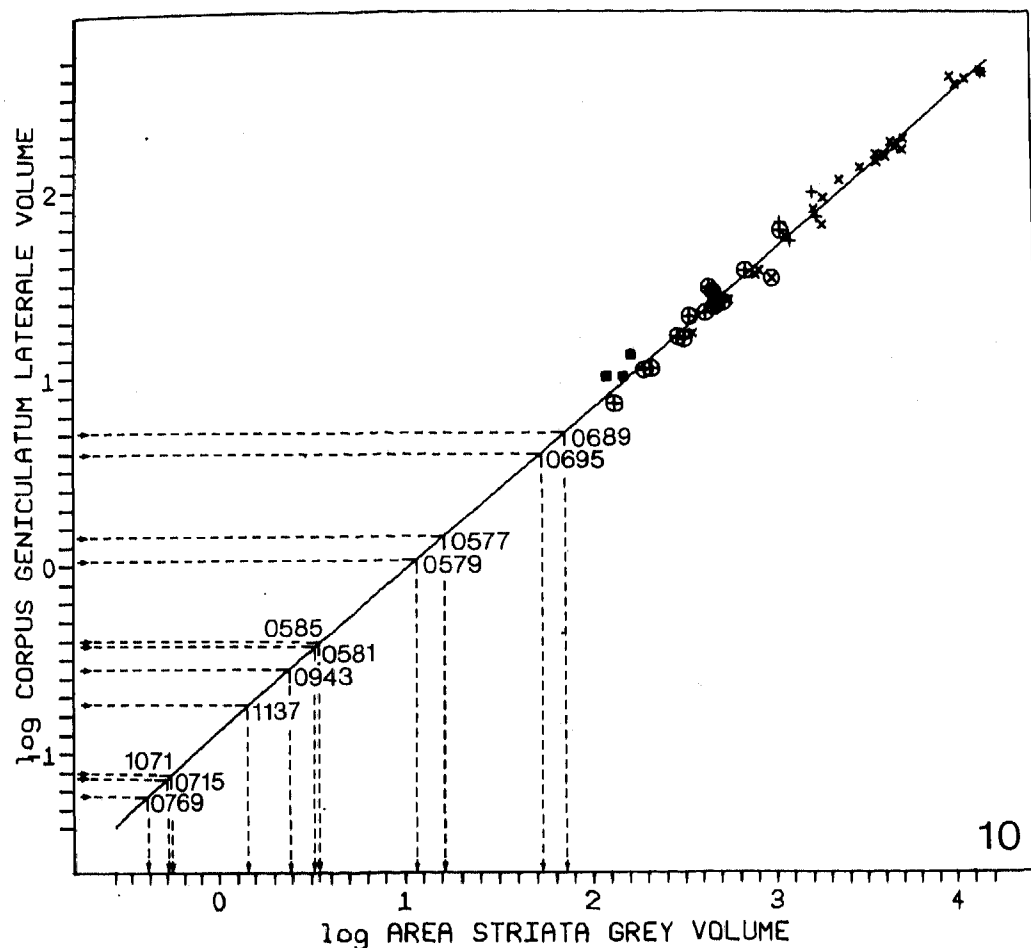


Fig. 10. Volume of the dorsal nucleus of corpus geniculatum laterale (CGL) plotted against volume of the area striata grey (ASG) in a double logarithmic scale. All volumes are in mm³. Symbols as in Figure 1. The formula for the regression line for the total of the 44 species of Primates and Scandentia investigated so far is:
 $\log \text{CGL} = -0.8801 + 0.8476 \cdot \log \text{ASG} \quad (r^2 \ 0.985).$

The CGL volumes of Insectivora are projected at right angles from the ordinate on the regression line and the corresponding ASG volumes read from the abscissa. The numbers close to the regression line are the code numbers of the Insectivora species as given in Table 4.

total neocortex (FRAHM et al., 1982), but it is less pronounced in ASG than in NG. This may be due to the fact that ASG1 is thinner than NG1. The average of ASG1 relative to ASG was found to be 13.6% (versus 15.2% for NG1 relative to NG) in prosimians, 10.6% (versus 12.8%) in simians and 11.9% (versus 13.8%) in all primates. A relatively smaller ASG1 stands out also when the ASG1 is calculated as a percentage of NG1 and compared with ASG in percentage of NG (cf. Table 2, columns 6 and 5).

Interstructural regression: When plotting ASG1 against NG1, the slope was 0.794 with a correlation coefficient (r^2) of 0.973. The values were similar to those of ASG versus NG. In a double logarithmic plot (Fig. 9) of ASG1 versus ASG, the overall regression line had a slope of 0.973 with a r^2 of 0.983. The slope, slightly smaller than 1, indicates

that with increasing ASG, the molecular layer increases somewhat less than the cell-rich layers 2–6.

Size indices: The average ASG1 index of prosimians was close to 100 (which is the reference basis), and the average of simians was 193. Thus the volume of the molecular layer of the primary visual cortex of simians was on the average 1.9 times larger than that of isoponderous prosimians. The index of man (229) was slightly above the simian average. The lowest ASG1 indices were found in the two tree shrews (*Urogale* 34; *Tupaia* 55). Within most families, the range of indices was very large (see Table 3, column 4; Fig. 5).

Laminae 2–6 of area striata grey (ASG2–6)

Percentage: The percentage values of the cell-rich cortical band (ASG2–6) relative to ASG complement those of ASG1 (see Table 2, column 9) bringing

them to 100. Thus the tendencies discussed for the percentage values of ASG1 are reversed: There is a slight percentage increase of ASG2-6 in simians, but there was great variability.

ASG2-6 in percentage of NG2-6 always gave slightly higher values than ASG in percentage of NG (cf. Table 2, columns 7 and 5). This compensates for the relatively lower percentages when relating ASG1 to NG1 (see above).

Interstructural regression: When plotting the volumes of ASG2-6 versus ASG in a double logarithmic scale (Fig. 9), the overall regression line had a slope of 1.004 and a correlation coefficient (r^2) of 0.9997. The very high correlation is not surprising since on the average, about 88% of ASG is ASG2-6. In a plot of ASG2-6 versus NG2-6, the slope was 0.798 with r^2 0.960. These values are nearly identical with those of ASG versus NG (see above).

Size indices: The ASG2-6 indices are given in Table 3, column 5 and scaled in Figure 6. The indices are only slightly different from the ASG indices and need not be discussed in detail (see Discussion of ASG).

Discussion

(1) Methods of comparison

Three methods were used to compare the size of the structures and each of the three provided instructive results when suitably applied.

Interstructural regressions were examined with regard to slopes and correlation coefficients. When, for example, the ASW is plotted against the ASG a very high slope indicates that with increasing visual cortex the white matter increases distinctly more than the grey matter; or when plotting ASG versus CGL, a high correlation coefficient points to a close size relationship between these two structures.

Percentage values are helpful, when, for example, the composition of the two laminar components of the visual grey investigated here (laminae 1 and 2-6) are compared. The percentages of the volumes of these laminar components relative to the total grey express their average relative thickness.

However, these first two methods cannot define clearly which of the two structures (related with one another) is responsible for the differences in inter-specific comparisons. A small percentage of ASG relative to NG, as found in man, or a low position distant from the interstructural regression line, does not necessarily imply that man has a small visual cortex, since this could also be the result of a large non-visual neocortex.

Size indices can give information about the comparative size of structural complexes in different species. The structures are plotted with the body weight and thus the size indices are nearly independent of the size of other brain structures. When plotting structure volume versus body weight, the slope of the reference base is of major importance and must be selected with great care to minimize differences in the size indices within narrowly related groups of species. In previous papers we have discussed the range of systematic groups (families, suborders, orders) across which the calculation of a common slope fulfills the requirement of minimizing differences in the size indices, and we will show that the found restrictions are valid also for the visual cortex. In the plot ASV versus BoW, regression analyses between families and primate suborders lead to slopes close to 0.53, whereas for the total of primate species a slope of 0.66 was found, and when the tree shrews are included, a value of 0.69 is obtained (see Fig. 1). When using a reference line with one of these steeper slopes, the differences of the size indices within narrowly related groups are not minimized and the smaller species would have larger indices than the larger ones. Thus, existing differences in relative AS size may be concealed.

In part IV (STEPHAN et al., 1984), we have shown that in all the plots of non-cortical visual structures versus body weight, the slopes are low and include the lowest so far found for brain structures. This is valid also for the primary visual cortex whose slopes (ASV 0.53; ASG 0.51) are similar to those of the optic tract (0.52) and CGL (0.48). These low slopes indicate that with increasing BoW, the increase of the visual structures is relatively weak.

(2) Comparative size of cortical and non-cortical visual structures

The artificial delimitation of the white matter does not allow for its exact measurement and may introduce errors into the measurements of the total AS. Therefore, we concentrate our discussion on grey matter comparisons. However, even when the size indices between white and grey matter differ considerably, as e.g. in *Lemur albifrons*, *Tarsius*, *Saguinus midas*, *Saimiri sciureus*, and *Miopithecus talapoin* (cf. Table 3, columns 2 and 3), the indices of the total AS (column 1) deviate little from that of the ASG and do not falsify the general results of the comparisons. Thus, comparisons of the total AS can also give useful results.

As the neocortex undergoes enlargement from lower to higher primates, there is an enormous increase of the white matter. This has been discussed in FRAHM et al. (1982). With all due reservations

(because of artificiality in the delimitations) the distinctly steeper slope of ASW compared with that of ASG (see Fig. 8) points in the same direction.

Scandentia: Tree shrews have smaller relative eye size than all prosimians and simians. In contrast, the indices for their optic nerves are distinctly larger than those of all prosimians and place them in the lower half of the simian range. The convergency from eye to the optic nerve (HSE/NOA ratio; see STEPHAN et al., 1984) is on the average the lowest of all the mammalian species so far investigated and place tree shrews among diurnal simians. The average CGL size of tree shrews is somewhat smaller than that of prosimians; however, there is complete overlap. Compared with simians, the relative CGL size is distinctly smaller. The size relations of ASG are similar to those of CGL, but the distances between tree shrews, prosimians and simians are somewhat larger.

The general trend in our size comparisons of the visual structures in tree shrews suggests: (1) overall weak development and (2) high adaptation to diurnal activities.

Prosimian primates, Cheirogaleidae: The indices of all the visual structures, both cortical and non-cortical, are smaller than the prosimian average. The high HSE/NOA convergency is in accordance with the nocturnal behaviour of this family of Madagascar.

Lemuridae: Within this family there are wide differences between the *Lepilemur* and the *Lemur* and *Varecia*. In the size of the visual cortex, these differences are marked by extreme positions within the total prosimian group. In *Lepilemur*, AS is smallest¹, whereas in *Lemur* and *Varecia*, it is largest. Similar extreme positions are found in the non-cortical visual structures, with the only exception being *Tarsius*, which has still larger HSE, NOA and CGL indices than *Lemur* and *Varecia*. The low indices of the visual structures in *Lepilemur* are in agreement with similar low indices found for the whole brain and many of its parts (STEPHAN, 1967, 1972). *Lepilemur* seems to have the lowest evolutionary level of all primates.

Lepilemur is a nocturnal species, whereas *Lemur* and *Varecia* are mainly or exclusively diurnal. The size indices of visual structures partly depend on

these differences in activity periods. The HSE/NOA convergency of *Lemur* and *Varecia* is the lowest within prosimians, and this is consistent with the general relationship between HSE/NOA convergency and day-activity.

Indriidae: As in Lemuridae, the nocturnal indrids (*Avahi*) are lower in relative CGL and ASG size than the diurnal *Propithecus* and *Indri*, but the differences are not as pronounced as in Lemuridae.

Daubentoniidae: In contrast to the very large brain of *Daubentonia*, which sets this species apart from all other prosimians (STEPHAN, 1967, 1972), all the visual structures, both cortical and non-cortical, are close to the prosimian average.

Lorisidae: The relative size of the visual cortex is not balanced within Lorisidae but there are some species with well-developed visual cortices (*Loris*, *Nycticebus*, *Galagoides*) and others with poorly-developed ones (*Perodicticus*, *Otolemur*). *Loris* approaches *Tarsius* in the relative size of its visual cortex. The results on visual cortex are in good agreement with those of CGL, whereas the eye (HSE) and optic nerve (NOA) behave differently (see STEPHAN et al., 1984). The poor development of visual structures in *Perodicticus* was already mentioned by SCHULTZ (1940) and SPATZ (1970).

Tarsiidae: In *Tarsius* one-fifth of the neocortex (about 21%, see also STEPHAN, 1984) is AS. This is the highest percentage within primates, but in size indices, *Tarsius* is surpassed by nearly all simians and within prosimians, by diurnal Lemuridae. CGL and NOA indices of *Tarsius* are largest in prosimians but fall within the range for simians, whereas the relative eye size is highest for all primates.

Tarsius is extremely well adapted to night-vision, but the HSE/NOA convergency is surprisingly low for a nocturnal species. There are yet other characteristics in which *Tarsius* leaves the range of nocturnal primates (*Aotus* included) and reaches into the range of the diurnal simians. This is valid for the NOA index, the "centralization" of the retina (ROHEN, 1958; CASTENHOLZ, 1984) and now, to some degree, also for the relative size of the visual cortex. The general impression is that *Tarsius* (1) has very large visual structures and (2) in several respects, approaches levels which in primates are otherwise restricted to diurnal simians.

Non-human simians: The only night active form, *Aotus*, has a low AS index and HASSLER (1966) pointed out that the differentiation of the layers is also weak. Still distinctly lower relative AS size is found in *Alouatta* and slightly lower size, in some marmosets (*Callithrix*, *Cebuella*). Relative sizes similar to

¹ The lowest position of *Lepilemur* in ASG indices induced STEPHAN (1969) to place the reference base through the values of this species (instead of through the prosimian average, as done here). In the present paper, the ASG index of *Lepilemur* based on the prosimian average is 53. To make the indices presented here comparable to those of our earlier work, the values given here must be multiplied by 1.89 (1:0.53).

those found in *Aotus* exist in *Colobus* and *Gorilla*. This is in agreement with the results on the relative size of the *non-cortical* visual structures (see STEPHAN et al., 1984) and confirms the general statements made in that report about low developed visual structures (1) in marmosets, (2) in the nocturnal *Aotus* (except large eyes), and (3) in leaf-eating or predominantly vegetarian species (*Alouatta*, *Colobus*, *Gorilla*) (see Discussion, section 5).

Man: The position of man in the comparative scale of relative AS size is not exceptionally high, but the indices are in general close to the average of simians. Similar positions resulted from a comparison of non-cortical visual structures and according to BONIN (1942), from the laminar cortex differentiation. Within the broad band of species investigated here, man seems to be the only one in which AS volume was measured by other authors too (GERHARDT and KREHT, 1933; M. ROSE, 1935; SAUER et al., 1983).

SAUER et al. (1983) measured the ASG volume in an ontogenetic series of brains (n 35) from 20-week old fetuses to 99-year old adults and found a clear relation between ASG size and age. Volumes equal to that of adults were already found in one year old children and the highest volume of all (20,700 mm³ in a 17-month old child. So far, volumes higher than 18,000 mm³ have been found only in children less than 14 years old. With increasing age, the ASG volumes show a clear tendency to decrease. From a regression analysis of SAUER et al., the following average volumes (years/ASG volumes of the two hemispheres in mm³) were obtained: 10/15,400; 20/14,800; 40/14,000; 60/13,100; 90/11,800. The value given in Table 1, column 4 (14,787 mm³) is from a 25-year old male and corresponds well with those given by SAUER et al.; their value would be about 14,600 mm³ for 25-year old individuals.

GERHARDT and KREHT (1933) measured in one human brain an ASG of 19,963 mm³ and in another of 9,918 mm³. No age were given for individuals, but N. POPOFF (1927) had previously measured the visual cortex areas in the same two brains, which were reported to be from adults. POPOFF arrived at large differences similar to those of GERHARDT and KREHT. He found 8,770 mm² for one AS and 4,199 mm² for the other. When compared with many other area measurements of the primary visual cortex (BRODMANN, 1911, 1913; TRAMER, 1914, 1916; ECONOMO and KOSKINAS, 1925; ECONOMO, 1926; PUTNAM, 1926; N. POPOFF, 1927; I. POPOFF and N. POPOFF, 1929; FILIMONOFF, 1932), the first brain deviates considerably from the results of all other investigations, which are between 2,300 and 4,914 mm². Thus, from the brains investigated by GERHARDT

and KREHT (1933), only the second seems to have a 'normal' AS size. As a correction factor, GERHARDT and KREHT added 20% (calculating the volume with a section thickness of 50 µm instead of 40 µm). Based on our experience, paraffin-embedded brains, which they used, in general undergo shrinkage of about 50%. After subtracting the 20% correction and, instead, doubling the volume, we reach about 15,870 mm³ grey matter in the AS of the second brain of GERHARDT and KREHT. This value corresponds fairly well with those of SAUER et al. and our own. Calculated in the same way, the second brain would have an ASG of nearly 32,000 mm³, which extends far beyond the range given by SAUER et al. (1983). ROSE (1935) found a value of 4,940 mm³ in one paraffin-imbedded hemisphere of an adult individual. If this value is doubled for the two hemispheres and again for shrinkage, a value of 19,760 mm³ results. This value is close to the upper border of the range given by SAUER et al., but the shrinkage of various brains can be highly different (see STEPHAN et al., 1981). However, from ROSE's data, the percentage of ASG relative to NG also is higher than in ours (3.3% versus 2.5%).

Both values (3.3% and 2.5%) are within the range found from *area* measurements, which are more numerous than volume measurements. The percentages found by various authors are as follows: BRODMANN (1911) about 3%; BRODMANN (1913) 2.65–2.9%; TRAMER (1916) 3.9%; ECONOMO and KOSKINAS (1925) 2–2.5%; ECONOMO (1926) 2.2%; I. POPOFF and N. POPOFF (1929) 2.8 and 4.0%. The values range from 2% to 4%, and have an average of 3%. Our volume measurements result in a somewhat lower value (2.5%), but such a difference seems to be more generally present in highly gyrencephalic brains. This can be shown in a comparison of the percentage values found by BRODMANN (1913) and I. POPOFF (1929) in higher primates with our own percentages from volume measurements (see next section).

(3) Area striata size in Insectivora

The presence of a visual field in at least several groups of Insectivora is indicated by both anatomical (BRODMANN, 1906, 1909; WATSON, 1907; M. ROSE, 1912; HALL and DIAMOND, 1968a; KUREPINA, 1974; SATO, 1977; REHKÄMPER, 1981) and physiological investigations (LENDE and SADLER, 1967; HALL and DIAMOND, 1968b; DIAMOND and HALL, 1969; ALLISON and VAN TWYVER, 1970; KAAS et al., 1970). Its presence in *moles* is doubted by BRODMANN, WATSON, ROSE and ALLISON and VAN TWYVER, but accepted by KUREPINA and SATO. SATO even investigated the relative size of supragranular and infragranular layers. For *shrews*, the existence of a visual

Table 4. Area striata grey (ASG) volumes estimated from the volumes of the dorsal nucleus of the lateral geniculate body (CGL). Ratios, percentages and size indices are based on the estimated ASG volumes given in column 2.

	Corpus gen. lat. CGL*)	Area striata grey ASG+)	Ratio ASG CGL	Percentage ASG NG§) · 100	Size indices of ASG§)
	mm ³ (1)	mm ³ (2)	(3)	(4)	(5)
0577 <i>Tenrec ecaudatus</i>	1.46	17.1	11.7	7.3	2.9
0579 <i>Setifer setosus</i>	1.05	11.6	11.0	9.9	3.7
0581 <i>Hemicentetes semispinosus</i>	0.367	3.35	9.1	5.0	1.6
0585 <i>Echinops telfairi</i>	0.373	3.41	9.1	7.4	1.8
0689 <i>Erinaceus algerus</i> ?)	5.02	73.3	14.6	16.8	12.9
0695 <i>Erinaceus europaeus</i>	3.90	54.4	13.9	12.0	9.2
0715 <i>Sorex araneus</i>	0.074	0.51	6.9	2.1	0.8
0769 <i>Sorex minutus</i>	0.059	0.39	6.6	3.3	0.9
0943 <i>Crociodura flavescens</i> §)	0.278	2.41	8.7	4.5	2.3
1071 <i>Crociodura russula</i>	0.077	0.53	6.9	2.2	0.8
1137 <i>Suncus murinus</i>	0.179	1.44	8.0	2.8	1.2
Means	n				
shrews	5		7.4	3.0	1.2
tenrecs	4		10.3	7.4	2.5
hedgehogs	2		14.3	14.4	11.0
tree shrews	2		13.6	14.4	65
prosimians	10		19.5	13.7	105
marmosets	2		22.8	16.5	190
monkeys	8		26.7	11.6	262
apes	2		29.1	8.0	235
man	1		35.6	2.5	274

*) CGL data taken from STEPHAN et al. (1984)

+) Estimated from $\log \text{CGL} = -0.8801 + 0.8476 \cdot \log \text{ASG}$ (see Figure 10)

§) NG data taken from FRAHM et al. (1982)

§) Calculated from $\log \text{ASG} = 1.277 + 0.51 \cdot \log \text{BoW}$ (see Table 3, column 3)

?) and §) See footnote of Table 1

field is generally accepted (WATSON, ROSE, RYZEN and CAMPBELL, KUREPINA, SATO), but the low differentiation has been stressed, whereas for *hedgehogs*, a better differentiation has been described (BRODMANN, WATSON, LENDE and SADLER, HALL and DIAMOND, DIAMOND and HALL, KAAS et al., REHKÄMPER). The differences found in anatomical and physiological investigations between various families of Insectivora confirm the results of comparative investigations on the non-cortical visual structures (STEPHAN et al., 1984).

When interstructural plots relating cortical with non-cortical visual structures (data of the latter taken from STEPHAN et al., 1984) are analysed, the strongest correlation is found between visual cortex and CGL. The regression line fits well for simians, prosimians and tree shrews and may also be valid for Insectivora. Therefore, the CGL versus ASG regression will be used here to estimate the size of the AS in Insectivora, in which order we were able to delimitate the CGL but not the visual cortex with

sufficient precision. To estimate ASG volumes, we projected the known CGL size of the Insectivora species onto the regression line (Fig. 10) and read the corresponding ASG size from the ordinate. The estimated volumes (given in Table 4, column 2) were then used to calculate percentages, ratios and indices for comparison with those of primates and tree shrews (given in Tables 2 and 3). The results of these calculations are included in Table 4. Even if we accept a tolerance of $\pm 100\%$ for the estimated volumes, many of the general trends, which are discussed below, do not change dramatically.

Based on the estimated volumes, ASG relative to NG is on the average 14.4% for hedgehogs, 7.4% for Tenrecinae and 3.0% for Soricidae. Within primates, a clear decrease of the percentage values from about 15% in prosimians and marmosets to 2.5% in man was found. This trend was not continuous to Insectivora, which in contrast to the expected higher values, in general had distinctly lower percentages. Only hedgehogs were at the level of tree

shrews, prosimians and marmosets, whereas tenrecs and shrews had low percentages similar to those of apes and man. However, similar low values in Insectivora and higher primates have quite different origins. While in higher primates, the enlargement of the non-visual cortex is the causal factor, in tenrecs and shrews the low values are due to the very low development of the visual structures.

In the European hedgehog, BRODMANN (1913) found from *area* measurements that 12.4% of the total neocortex is visual cortex. From a comparison of the estimated visual cortex *volume* with total neocortex volume, a similar percentage (12.0) results (Table 4, column 3). The question arises whether this agreement between the results of area and volume measurements can be regarded as a confirmation of our volume estimation, or whether it is accidental. BRODMANN (1913) and I. POPOFF (1929) made area measurements on the visual cortex. They gave the following percentages of AS surface relative to neocortex surface from primate species, which are in part included in our material, or in part closely related to species investigated here. The percentages are (our data in brackets): *Lemur* 13.2 (13.7), *Cheirogaleus* 21.1 (20.2), *Callithrix* 22.3 (15.9), average of monkeys 13.3 (11.6), gibbon 12.2 (9.8), chimpanzee 8.2 and 8.3 (6.1) and man 2.7 (2.5). In simians, the area percentages given by BRODMANN and I. POPOFF are in general somewhat higher than our volume percentages, whereas in prosimians, they are similar. The differences may be due to the different degree of neocortical fissuration. The neocortex of Insectivora is expected to behave more similarly to that of prosimians. We expect that in Insectivora and prosimians the percentage of visual cortex obtained from area measurements would be close to that obtained from volume measurements. Then, BRODMANN's value from the hedgehog would confirm the estimated volume and, accordingly, our method of estimation at least for this relatively large species. For the smaller species, we have so far had no confirmation.

The ratio ASG/CGL increases from shrews through tenrecs to hedgehogs. Based on the low slope in the interstructural regression CGL versus ASG (Fig. 10), species with small CGL volumes (mainly small-bodied species) have relatively smaller ASG volumes and correspondingly ASG/CGL ratios than species with large CGL volumes (mainly large bodied species). The tendency of the ratios to decrease from man (35.6) to prosimians (19.5) continues in Insectivora (Table 4, column 3).

The size indices obtained from the estimated ASG volumes are lower than those found for CGL (STEPHAN et al., 1984). The average ASG indices (CGL indices in brackets) are for: Erinaceidae 11.0 (15.0),

Tenrecinae 2.5 (4.5), and Soricidae 1.2 (2.8). According to these indices, the visual cortex of shrews is about 1/90, that of tenrecs about 1/40, and that of hedgehogs about 1/10 that of isoponderous prosimians.

(4) Laminar differentiation

According to STEPHAN (1969), the thickness of ASG is remarkably constant, independent of differences in body and/or brain size. The slope of the regression line is 0.09 only, and an enlargement of the cortical surface seems to be primarily responsible for the increasing volume of the visual cortex. From our volume measurements of laminae 1 and 2–6, the average relative thickness of these laminar components can be determined. The results point (1) to a small molecular layer of the visual cortex relative to that of the neocortex as a whole (11.9% versus 13.8%), and (2) to a slight decrease of the percentage of lamina 1 relative to ASG when prosimians are compared with simians (13.6% versus 10.6%). This decrease is in agreement with a similar decrease of the molecular layer of the total neocortex from Insectivora (28.3%) through prosimians (15.2%) to simians (12.8%) (see Discussion in FRAHM et al., 1982).

Measurements on thickness in rodents indicate a lower percentage of the molecular layer in the brain of the arboreal squirrel, in which the visual cortex is much better developed, than in mouse, rat, and guinea-pig, which are terrestrial forms (VOLKMANN, 1928; GUREWITSCH et al., 1929; POPOVA, 1959; WINKELMANN et al., 1972).

The quantitative differences between molecular layers of the visual cortex and the whole cortex as well as the reduction of the molecular layer from Insectivora to simians are paralleled by differences in the laminar organization of the commissural connections. According to experimental studies of HEIMER et al. (1967; rat, opossum) confirming earlier descriptions of VILLAVERDE (1932a, b; guinea pig, mouse) commissural fibers end in several layers but mainly in the superficial layers 1 and 2. There are, however, regional differences in the relative number of callosal fiber terminals. Of particular interest is that in area 17, for example, few degenerating axons have been identified by the authors. It is possible that the low number of callosal terminals may account for the reduced thickness of the molecular layer in this area.

A comparison of the sites of commissural fiber endings in four species including rhesus monkeys and chimpanzees (JACOBSON and MARCUS, 1970) has shown the existence of large interspecific variation in the laminar termination of the callosal system. Whereas numerous terminals have been found in layers 1 and 2 of the rat and the cat, there are no such terminals in layers 1 and 2 of primates. In pri-

mates, the granular and not the supragranular layer is the major receptive layer for callosal input (LORENTE DE NÓ, 1949; JACOBSON and MARCUS, 1970). These findings suggest that the reduction of the molecular layer in the ascending primate scale is at least partly due to a reduction of the callosal fibers terminating in this layer.

ZILLES et al. (1982) stated that with increasing ASG index, the grey-level-index (GLI) also increases. The nerve cells in the granular layer become more densely packed, and this is evidence for the generally increasing granularization from lower to higher neocorticalized mammals. In contrast, the supragranular layer of the visual cortex shows a tendency for a decreasing GLI. This indicates increasing neuropil volume and thus also shows a trend towards progressive development. According to ZILLES et al. (1982) the species differences can be related, in part, to functional adaptations and, in part, to evolutionary processes.

(5) *Ecoethological aspects*

Since vision varies under different environmental circumstances, it is reasonable to interpret the adaptive nature of quantitative differences in visual brain structures in the light of ecological and behavioural data.

Activity periods

One of the most conspicuous aspects of the ecological niche to which vision is adapted is the time of an animal's active period. The relative size of the visual cortex is lower in nocturnal species than in closely related diurnal species. The mean size index (ASG) for nocturnal prosimians was 98 (n 14), that for diurnal, 128 (n 4). The coefficient of variation (CV) is similar in both cases (31% and 29% respectively). Among simians, the size index of the only nocturnal species was 171 whereas the mean index of diurnal species reached 250 (CV 28%, n 22).

Similar relations are found in the non-cortical visual structures. CGL is relatively smaller in nocturnal than in closely related diurnal species, and according to HASSLER (1965, 1966) and SCHULZ (1967), the nocturnal primates have a distinctly higher percentage of the magnocellular components, both in volume of the layers and in their cell numbers. Especially clear differences exist in the HSE/NOA convergency, i.e. from the retina to the optic nerve, which in nocturnal species is distinctly stronger than in diurnal forms. Relatively large eyes combined with relatively small optic nerves are typical of nocturnal primates. ROHEN and CASTENHOLZ (1967) found that

in general the "centralization" of the retina (ROHEN, 1958) is lower in nocturnal primates than in diurnal ones (except *Tarsius*, see section on Tarsiidae).

Feeding categories

A classification of the species investigated here according to their foraging techniques was used by BARON et al. (1983) in their study of the relative size of the main olfactory bulb. This classification comprised three dietary grades and was based on investigations by HLADIK (1975, 1979). With respect to diet preferences, HLADIK (op. cit.) distinguishes three grades: insectivorous and carnivorous species complementing their diet with fruits and gum (grade 1); frugivorous forms, which also include gum- and seed-eaters that eat a certain amount of animal prey (grade 2); and finally, folivores, which eat shoots and flowers with a supplement of various amounts of fruit in addition to leaves (grade 3). Contrary to the claim of BARON et al. (1983) that *Cercopithecus mitis* and *C. ascanius* may be classified with grade 3, it now seems that both species are mainly fruit-eaters (CLUTTON-BROCK and HARVEY, 1977). In fact they do not differ from the mainly frugivorous species in terms of the development of their olfactory (BARON et al., 1983) and visual brain structures.

Prosimians: Differences in feeding ecology are strongly reflected by differences in the development of the visual system, especially the area striata. Folivorous prosimians, with a mean index of ASG of 101 (CV 43%, n 7), have the largest range of variation among prosimian dietary types indicating that there must be marked differences in their feeding strategies. Diurnal folivorous species attain the highest mean index among prosimians ($\bar{x}$ 128, CV 29%, n 4), whereas nocturnal leaf-eaters have the least developed striate area among primates ($\bar{x}$ 65, CV 18%, n 3). But even among diurnal folivores, the range of variation is still high. Thus, the ASG has a mean index of 96 among Indridae and a mean index of 160 among Lemuridae. There are important differences in the behavioural ecology of these groups. According to HLADIK (1979), *Lemur* can more easily adapt to changes in diet than, for instance, *Indri* and *Propithecus*, which are more specialized in their natural diets. There are also differences in positional behaviour. When leaping from trunk to trunk Indridae maintain a more or less vertical posture, usually following a flat trajectory (PETTER, 1962; PETTER et al., 1971; WALKER, 1979). The estimation of distance is, in such cases, relatively unimportant (PARIENTE, 1979). In Lemuridae, the body posture during leaps is more inclined in the direction of the leap. Lemuri-

dae also perform leaps from a higher to a lower level which demand precise evaluation of the distance (WALKER, 1979; PARIENTE, 1979).

In nocturnal leaf-eaters (*Lepilemur* and *Avahi*), the low indices of the visual system are not compensated by a better developed olfactory bulb (BARON et al., 1983). They differ from other folivorous species by their particular feeding strategy, feeding mainly on the tough foliage of various trees commonly found in the forest (CHARLES-DOMINIQUE and HLADIK, 1971; HLADIK, 1979). They belong to what RIPLEY (1979) calls the strict folivores. *Lepilemur*, which occupy a relatively small home range and show low dietary diversity (CLUTTON-BROCK and HARVEY, 1977), need to process less varied visual information.

All prosimians belonging to feeding category "grade 2" (see above) are nocturnal forms. The mean size of the area striata ($\bar{x}$ 88, CV 19%, n 5) is larger in this group than in nocturnal folivores. Frugivores tend to have larger home ranges than folivores (CLUTTON-BROCK and HARVEY, 1977). Since fruit trees are usually widely dispersed or have a patchy distribution, frugivorous species have to travel over ranges exceeding those of leaf-eaters of similar size. For instance, the distance covered by *Galago senegalensis* each night is about 2 km (CHARLES-DOMINIQUE and BEARDER, 1979). Locomotion and orientation in a complex environment certainly require efficient nocturnal vision, and thus visual discrimination and visual learning capacities should be better developed in fruit-eaters than in leaf-eaters to integrate the greater variety of visual information. There is good evidence that sight is important in learning about features of the environment (CHARLES-DOMINIQUE, 1971; CHARLES-DOMINIQUE and BEARDER, 1979).

Among the night-active prosimians, insectivorous and carnivorous forms (grade 1) have the best developed area striata (ASG; $\bar{x}$ 124, CV 20%, n 6). *Tarsius*, at the top of the grade 1 scale, is the only primate which feeds exclusively on animal prey comprising invertebrates and small vertebrates (HLADIK, 1979; NIEMITZ, 1979). According to NIEMITZ (1979), tarsiers locate their prey mainly by sound and sight.

Microcebus murinus and *Galagoides demidoff* resemble each other in terms of diet, feeding strategy and ecology (HLADIK, 1979). They feed mainly on insects supplemented by various amounts of birds, fruits and gums (CHARLES-DOMINIQUE, 1974, 1977). There are, however, differences in their respective micro-environment. *Microcebus* live in the dense undergrowth, *Galagoides* live at the top of the canopy. Both species hunt large fast-moving insects by stereotyped grabbing movements (HLADIK, 1979). *Loris tardigradus* and *Nycticebus coucang* are different from *Microcebus* and

Galagoides in terms of food. They catch prey which are generally avoided by other sympatric insect-eaters because of their repugnant smell (ELLIOT and ELLIOT, 1967; HLADIK, 1979). Despite the predominance of olfaction in the detection of prey (PETTER and HLADIK, 1970; HLADIK, 1975; BARON et al., 1983) vision is an important source of information about the environment. The size indices of the striate area in *Loris* and *Nycticebus* are the highest among night-active prosimians, except *Tarsius*.

Simians: The visual system in simians differs from that in prosimians in several details. Most prosimians are nocturnal but only one simian (*Aotus trivirgatus*) is known to have scotopic vision. Colour vision in diurnal prosimians is less developed than in simians (PARIENTE, 1979). Finally, as our results show, the visual brain centers, especially the area striata (ASG), are better developed in simians ($\bar{x}$ 247, CV 29%, n 23) than in prosimians ($\bar{x}$ 105, CV 32%, n 18). Despite the good vision in prosimians, there is no doubt that visual cues assume an increasingly important role in simians as compared to prosimians. Improved vision provides richer information about the ecological niche and facilitates food detection, foraging techniques, and food manipulation. In fact, simians execute much more versatile finger movements than prosimians; this would be impossible without their excellent vision.

In addition to this tendency of size to increase from prosimians to simians, there are other aspects of the quantitative analysis deserving special attention. Despite the apparently greater uniformity in visual sensitivity (only one nocturnal species) and in dietary types (no grade 1 and only few folivores), the degree of variability is only slightly smaller in simians (CV 29%) than in prosimians (CV 32%). Therefore the existing quantitative differences must also be related to other factors in the niche occupied.

Variations in the complexity of the habitat, home range size and the height and other characteristics of the forest stratum, as well as their separation into terrestrial and arboreal types, into various positional and locomotory modes, and into different grades of social structure, all contribute to the diversification of niches and determine the complexity of visual information gathered in order to produce the appropriate behavioural response.

In contrast to the case of the main olfactory bulb (BARON et al., 1983), among the diurnal neotropical simians, Callitrichidae have on the average a smaller visual cortex (ASG; $\bar{x}$ 190, CV 18%, n 4) than do Cebidae ($\bar{x}$ 250, CV 36%, n 7). The difference is still greater if only those species which belong to grade 2 are compared (that is *Alouatta* excluded). In this case, Cebidae have a mean index of 272 (CV 28%, n 6).

These differences may be related to differences in the size of the home range. Cebidae generally occupy larger home ranges than do Callitrichidae. According to CLUTTON-BROCK and HARVEY (1980) the mean home range size of Cebidae is 0.210 km^2 whereas that of Callitrichidae is about 0.150 km^2 . It may be assumed that the former need to store a wider range of visual information than the latter.

Alouatta, which is predominantly a leaf-eater (HLADIK, 1975), has the lowest ASG index (116) among Cebidae, even lower than the night-active *Aotus* (174). The same trend is observed among paleotropical simians. Folivores have the least developed area striata in their respective families. Thus *Colobus* has an index of 177 whereas the frugivorous Cercopithecinae reach a mean index of 306 (CV 11%, $n = 6$). Similarly, the index of *Gorilla* (175) is much lower than the mean index of the two frugivorous Pongidae (*Pan* and *Hylobates*) used in our study (265, CV 16%).

The low relative ASG size in the leaf-eating species is consistent with the low relative size of the non-cortical visual structures (STEPHAN et al., 1984), as well as with other brain structures and the total brain (STEPHAN, 1967, 1972). There seems to be a relationship between low relative size of the brain and its parts, on the one hand, and a leaf-diet, on the other (BAUCHOT and STEPHAN, 1969; CLUTTON-BROCK and HARVEY, 1980). The generally low relative size of the brain structures points to the possibility that not the brain, but rather a disproportionally large body size, is the releasing factor. We have discussed this possibility several times with reference to the very low position of the *Gorilla* (e.g. FRAHM et al., 1982). Like herbivores, specialized folivores need larger digestive organs and the food needs more time to pass through the body. For these reasons, the body size may be larger in folivorous than in carnivorous and frugivorous species and thus, their relative brain size is smaller.

Similarly, disproportionally small body size (dwarfism) may be the releasing factor, when all the brain parts are relatively large, as e.g. in *Miopithecus talapoin*. *Miopithecus* had the third highest ASG index and the highest CGL index of all species investigated. Such generally high relative size should not be interpreted only in functional terms, but must also take into account the relatively small body size. However, species with reduced body size, and thus large relative brain size, are exceptions in an otherwise strong relationship between body size and size of brain and/or brain components.

There are also, however, satisfactory functional explanations for the differential growth of visual structures in folivores despite the particular relation-

ship between body size and brain size mentioned above. As was already seen in the case of folivorous prosimians, folivorous simians in general occupy smaller home ranges with more predictable food supplies than do the frugivorous forms (CLUTTON-BROCK and HARVEY, 1977).

Comparison with cytoarchitectonic variations

The adaptive value of size differences becomes still more obvious when they are related to cytoarchitectonical characteristics. ZILLES et al. (1982) described the relative width of the layers of the striate area and found wider supragranular layers in diurnal species but wider granular layers in nocturnal species.

Among the nocturnal forms, there is a marked trend toward increased development of the granular layer in insectivorous (grade 1) species in parallel to a decreased supragranular layer. *Tarsius* has the largest relative width of the granular layer, and together with *Microcebus*, the smallest relative width of the supragranular layers. In contrast, the frugivorous (grade 2) *Cheirogaleus* has the relatively largest supragranular layers and the relatively smallest granular layer. The folivorous *Lepilemur* and *Avahi* occupy intermediate positions with respect to the relative width of the cortex layers.

According to ZILLES et al., the relatively large visual cortex of *Tarsius* is not really much advanced when compared with that of other primates, since, because of the relatively larger granular layer, it is engaged primarily in information perception and less in higher information processing by intercortical association, a role assumed mainly by the supracortical layers.

Quantitative variations may be explained by differences in the micro-environments in addition to differences in feeding strategies. Light in particular differs in composition and intensity owing to differential absorption by the biotype at different heights in the forest.

According to PARIENTE (1980), nocturnal prosimians are particularly sensitive to wavelengths of 450 to 500 nm indicating that they are particularly adapted to utilize contrast cues for detecting objects illuminated by the blue-green zenithal light against the more reddish forest background. In this context, differences in the behavioural ecology of *Microcebus* and *Cheirogaleus* may be of special interest. There is, between the two sympatric species, a significant difference in the mean heights above the ground where the animals were usually observed; like *Tarsius*, *Microcebus* occupy the lower strata in the forest. *Microcebus* and *Cheirogaleus* differ too in their distribution and feeding activity during the night (HLA-

DIK et al., 1980). This indicates that they prefer different spectral environments.

One may assume that the use of contrast cues is particularly important for species such as *Microcebus*, *Tarsius* and other grade 1 prosimians which hunt prey in the night under poor light conditions. The opposite positions that *Microcebus* and *Cheirogaleus* occupy with respect to the laminar composition of the striate cortex despite similar size indices may be related to the above-mentioned differences in their micro-environments.

On the grounds of the previous discussions, one may assume that the size index of the area striata indicates its general phylogenetic progress, whereas the relative laminar composition seems to reflect the particular direction of the functional adaptations.

(6) Phylogenetic aspects

The neocortex has proved to be the most reliable indicator for evaluating evolutionary progress. A comparison of the relative size (indices) of the area striata with those of the neocortex (the latter taken from FRAHM et al., 1982) reveals that from Insectivora to prosimians and from prosimians to non-human simians, the increase of the visual cortex was about the same as that of the total neocortex. When related to the estimated size in the hedgehogs, whose extinct relatives are considered as forerunners of primates, the visual cortex of the average prosimian has enlarged about 10 times and the neocortex as a whole (also based on hedgehogs), about 11 times. The further enlargement from prosimians to non-human simians is 2.3 times for the visual cortex and 2.2 times for the total neocortex. These values indicate that the development of the visual system to become the main sensory system in primates obviously took place in the early phase of primate development from Insectivora to prosimians. In the phase from prosimians to simians, a further consolidation of vision within the enlarging neocortex may have occurred. In the late phase of *human* development, the further enlargement of non-visual cortices predominated, whereas the relative size of visual structures remained on a level already reached by monkeys. These marked differences explain the strong reduction of the percentages of the visual cortex relative to total neocortex in man. Although the human visual cortex is distinctly progressive and enlarges 2.7 times as compared to that of isoponderous average prosimians, by no means representing a reduction, its percentage size decreases dramatically from 13.7% in prosimians to 2.5% in man. Long ago, BRODMANN (1906) noted that the percentage of the

visual cortex relative to neocortex is larger in lower gyrencephalic primates than in man.

Our results indicate a predominating influence of functional adaptations on the visual structures, whereas the influence of general evolutionary progress appears to be minor. This applies more to eyes and the optic nerves than to the CGL and visual cortex, where there is greater evidence of an influence of general evolutionary progress, e.g. increasing neocorticalization. This interpretation is confirmed by the investigations of ZILLES et al. (1982) on the comparative structural differentiation of the visual cortex (see above).

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